

SOURCE CODE:

```
#include <stdio.h>

#include <time.h>


#define MAX 1000


/* ----- Merge Sort ----- */


void merge(int arr[], int l, int m, int r)
{
    int i,j,k;

    int n1=m-l+1;

    int n2=r-m;

    int L[MAX],R[MAX];

    for(i=0;i<n1;i++)
        L[i]=arr[l+i];

    for(j=0;j<n2;j++)
        R[j]=arr[m+1+j];

    i=0;

    j=0;

    k=l;
```

```
while(i<n1 && j<n2)
{
    if(L[i]<=R[j])
        arr[k++]=L[i++];
    else
        arr[k++]=R[j++];
}
```

```
while(i<n1)
    arr[k++]=L[i++];
```

```
while(j<n2)
    arr[k++]=R[j++];
}
```

```
void mergeSort(int arr[],int l,int r)
{
    if(l<r)
    {
        int m=(l+r)/2;

        mergeSort(arr,l,m);
        mergeSort(arr,m+1,r);
```

```

        merge(arr,l,m,r);
    }
}

/* ----- Quick Sort ----- */

int partition(int arr[],int low,int high)
{
    int pivot=arr[high];
    int i=low-1;
    int j,temp;

    for(j=low;j<high;j++)
    {
        if(arr[j]<pivot)
        {
            i++;

            temp=arr[i];
            arr[i]=arr[j];
            arr[j]=temp;
        }
    }

    temp=arr[i+1];

```

```

arr[i+1]=arr[high];
arr[high]=temp;

return i+1;
}

void quickSort(int arr[],int low,int high)
{
    if(low<high)
    {
        int pi=partition(arr,low,high);

        quickSort(arr,low,pi-1);
        quickSort(arr,pi+1,high);
    }
}

/* ----- Main ----- */

int main()
{
    int n,i;

    int arr1[MAX];
    int arr2[MAX];

```

```
clock_t start,end;
```

```
printf("Enter number of elements: ");
```

```
scanf("%d",&n);
```

```
printf("Enter elements:\n");
```

```
for(i=0;i<n;i++)
```

```
{
```

```
    scanf("%d",&arr1[i]);
```

```
    arr2[i]=arr1[i];
```

```
}
```

```
start=clock();
```

```
mergeSort(arr1,0,n-1);
```

```
end=clock();
```

```
double mergeTime=((double)(end-start))/CLOCKS_PER_SEC;
```

```
start=clock();
```

```
quickSort(arr2,0,n-1);
```

```
end=clock();
```

```
double quickTime=((double)(end-start))/CLOCKS_PER_SEC;
```

```
printf("\nMerge Sort Output:\n");

for(i=0;i<n;i++)
    printf("%d ",arr1[i]);

printf("\nExecution Time = %lf seconds\n",mergeTime);

printf("\nQuick Sort Output:\n");

for(i=0;i<n;i++)
    printf("%d ",arr2[i]);

printf("\nExecution Time = %lf seconds\n",quickTime);

return 0;
}
```